

# Computer Science & IT

## Discrete Mathematics



### Graph Theory

Lecture No. 09



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# Recap of Previous Lecture



- ✓ **Topic** Vertex covering
- ✓ **Topic** Vertex independent set
- ✓ **Topic** Connectivity



# Topics to be Covered



✓ Topic

Connectivity

✓ Topic

Spanning tree

✓ Topic

Euler trail and Euler circuit

✓ Topic

Hamiltonian path and Hamiltonian cycle

✓ Topic

Distance, Eccentricity, Diameter, Radius, Girth

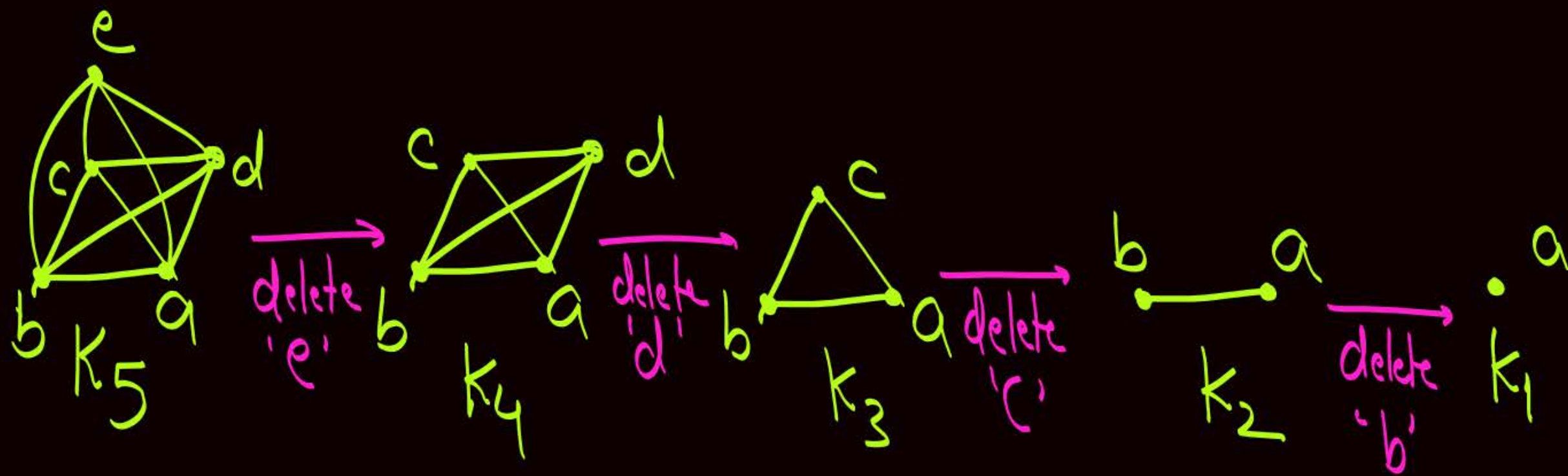
✓ Topic

n-cube graph

Q:- Find  $\lambda(G)$  and  $K(G)$  for Complete graph  $K_n$ .

•  $\lambda(K_n) = (n-1)$

$K(K_n) = (n-1)$





Note: If an isolated vertex exists along with other components in the graph, then isolated vertex is treated as a Connected Component

- If there is only '1' isolated vertex then it is Exceptional Case in which it may be treated as Connected or disconnected as per situation

H.W. Find Vertex Connectivity ( $K(G)$ ) and Edge Connectivity ( $\lambda(G)$ )  
For the following graphs.

|                                      | Vertex Connectivity                      | Edge Connectivity |
|--------------------------------------|------------------------------------------|-------------------|
| ① Complete graph $K_n$               | $(n-1)$                                  | $(n-1)$           |
| ② Cycle graph $C_n$                  | 2                                        | 2                 |
| ③ Wheel graph $W_n$                  | $1 + 2 = 3$<br>Hub $\nearrow$ from cycle | 3                 |
| ④ Complete bipartite graph $K_{m,n}$ | $\min(m, n)$                             | $\min(m, n)$      |
| ⑤ Star graph with $n$ -vertices      | 1                                        | 1                 |



#Q. In a connected graph G if we delete an edge then no. of components are

**A** 0 or 1

☒ **B** 1 or 2

**C** 2

**D**  $\geq 2$

[MCQ]

with 'n' vertices

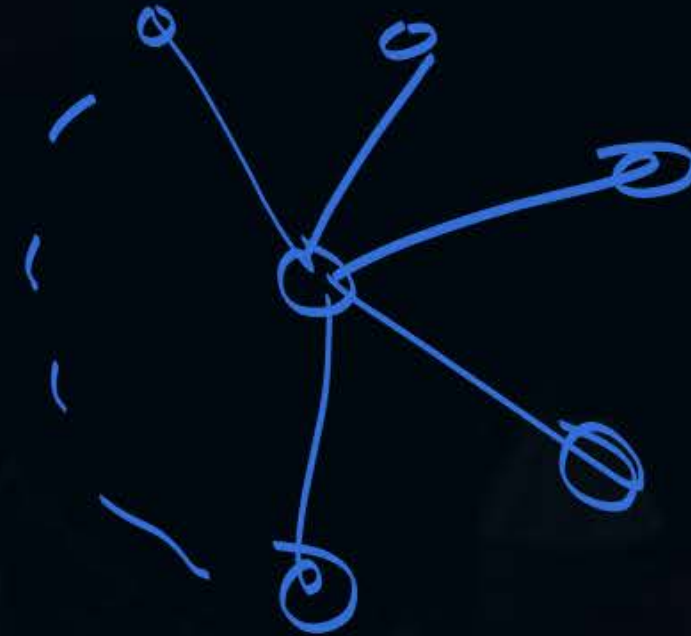
#Q. In a connected graph G if we delete a vertex then number of components are

**A**  $\leq 2$

**B** 1 or 2

☒ **C** Lies between 1 and  $n - 1$

**D**  $\geq 2$





H.W.



#Q. A simple graph with  $n$  vertices is necessarily connected if number of edges are more than  $E$ , then find the value of  $E$ .

•  
isolated vertex

Complete graph with  $(n-1)$  vertices &  $(n-1)C_2$  edges

It is a disconnected graph with ' $n$ ' vertices &  $(n-1)C_2$  edges

But if  $\# \text{Edges} > (n-1)C_2$  in a simple graph then it is necessarily connected  
 $\therefore E = (n-1)C_2 = \frac{(n-1)(n-2)}{2}$



H.W.



#Q. Maximum number of edges possible in a simple undirected graph with  $n$ -vertices and  $k$  components?



Complete graph  
with  $(n-k+1)$   
vertices &  
 $\binom{n-k+1}{2}$  edges

$$\begin{aligned} \therefore \text{Maximum no. of} \\ \text{Edges possible} \\ &= \binom{n-k+1}{2} \\ &= \frac{(n-k+1)(n-k)}{2} \end{aligned}$$

Upto  $\binom{n-k+1}{2}$  edges, it may be possible to have a simple graph with ' $n$ ' vertices & ' $k$ ' components, but if no. of Edges are more than  $\binom{n-k+1}{2}$ , then ' $k$ ' components are not possible



Note:

A simple graph with 'n' vertices & 'k' components must have at least  $(n-k)$  edges and can have at most  $(n-k+1)$  edges.

Q:- Let  $G$  be a graph with ' $n$ ' vertices & ' $k$ ' Components, if we delete a vertex from graph  $G$  then no. of Components can lie in the range from  $x$  to  $y$

$x = (k-1)$  { When vertex being deleted is an isolated vertex }

$y = (n-1)$  { w.r.t. Star graph }



Spanning Tree



## Topic : Spanning Tree



Let  $G$  be a connected graph,

If graph is not a Connected graph, then spanning tree does not exist for that graph

A sub-graph  $H$  of graph  $G$  is called a spanning tree if and only if

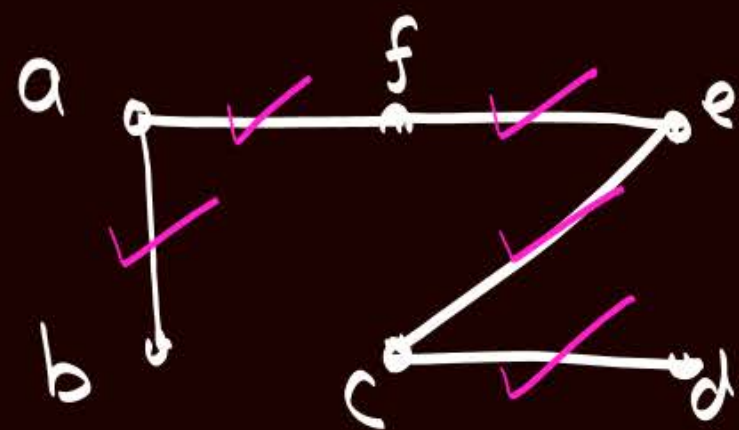
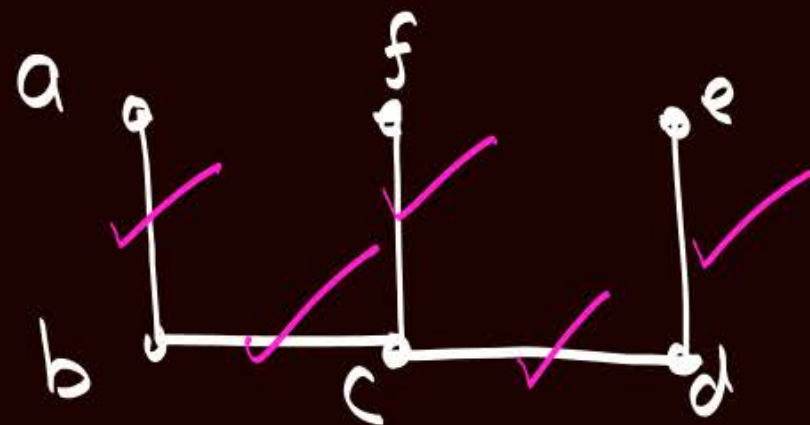
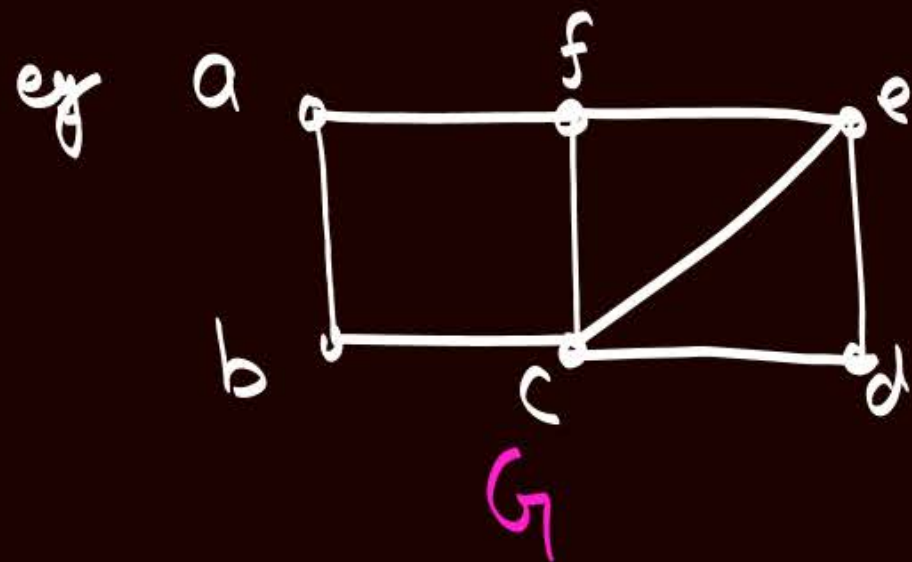
- $H$  is a tree (Acyclic + Connected)

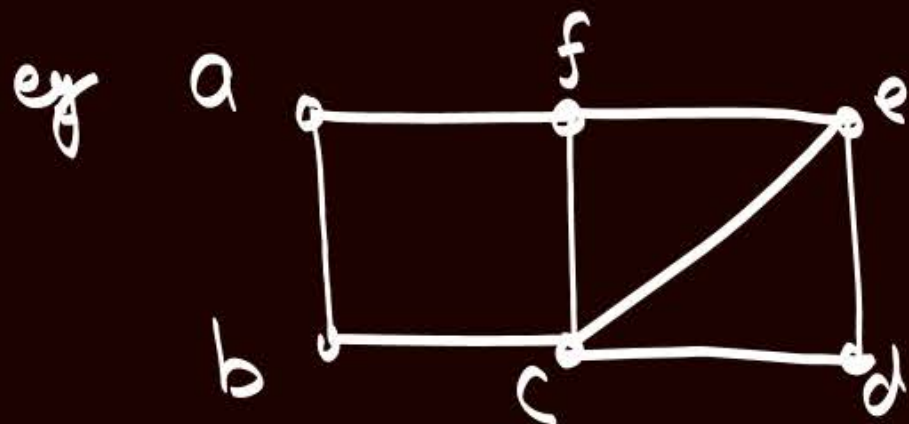
and

- $H$  contains all vertices of graph  $G$

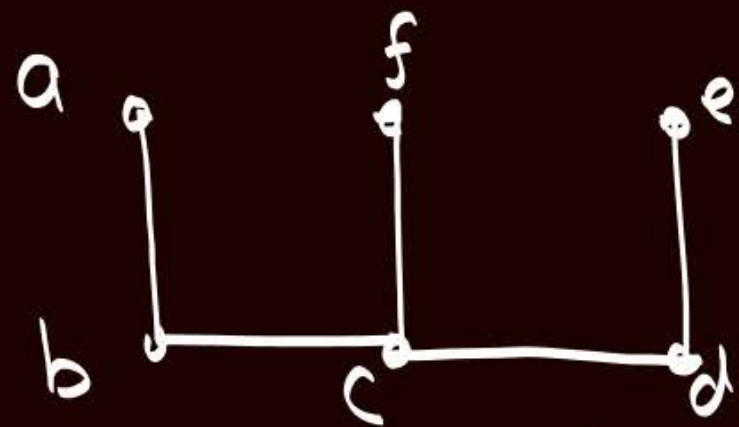
A graph  $G$  is connected  
if and only if  
Spanning tree exist for graph  $G$







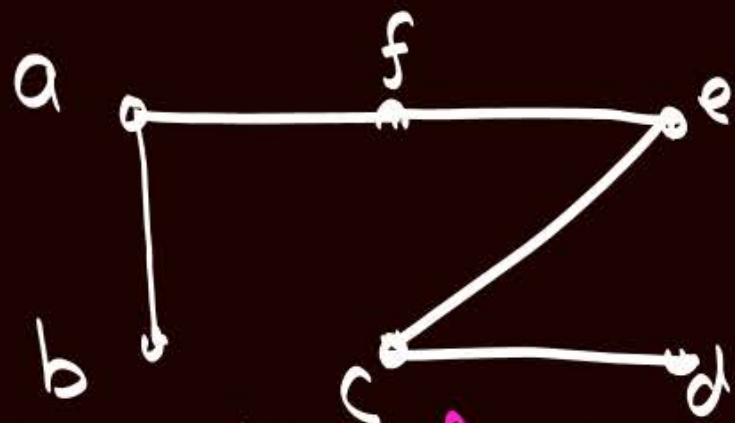
$G$



Sub-graph  $H_1$  of graph  $G$

Acyclic ✓ + Connected ✓ + Contain all vertices ✓  
tree

$\therefore H_1$  is a spanning tree of  $G$

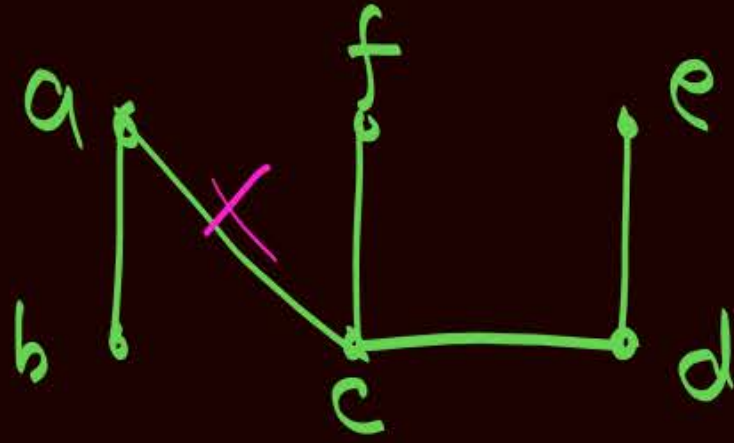
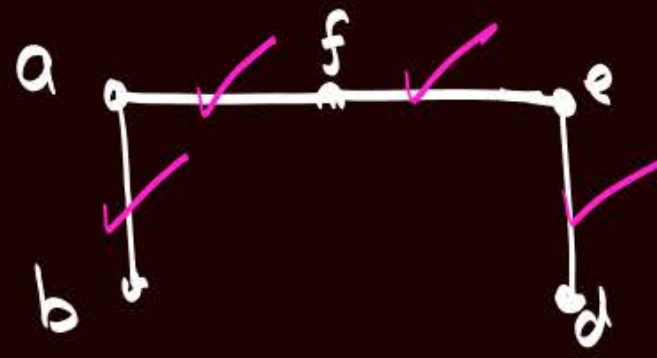
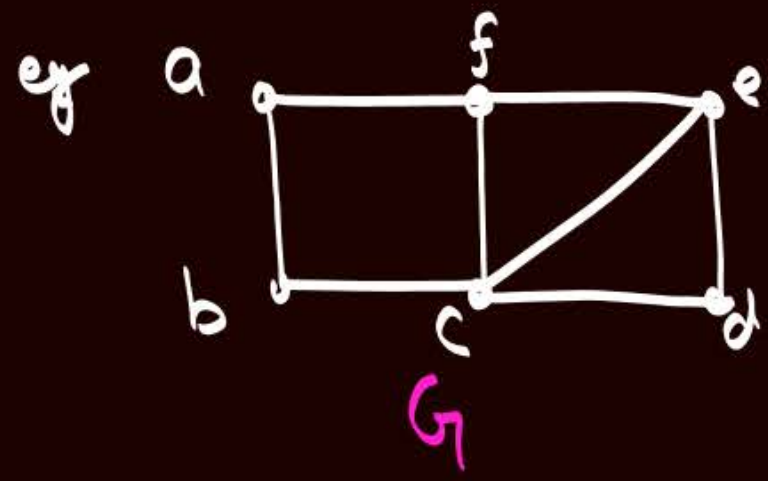


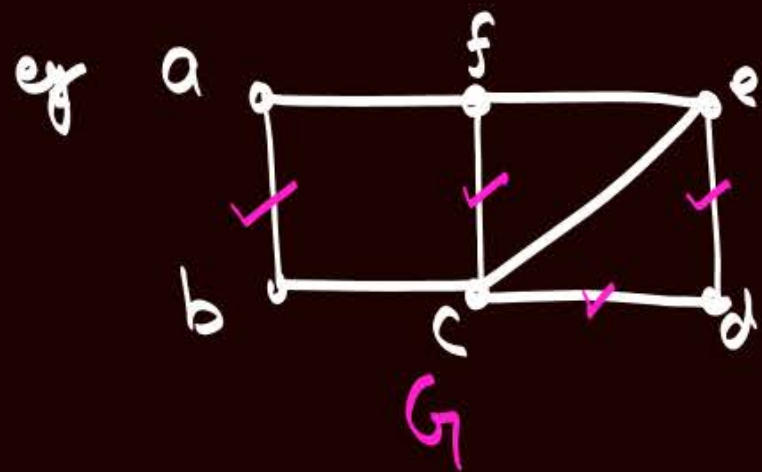
Sub-graph  $H_2$  of graph  $G$

Acyclic ✓ + Connected ✓ + Contain all vertices ✓  
 $\therefore H_2$  is a spanning tree of  $G$

For a connect graph  $G$ ,  
Spanning tree of  $G$   
need not be unique

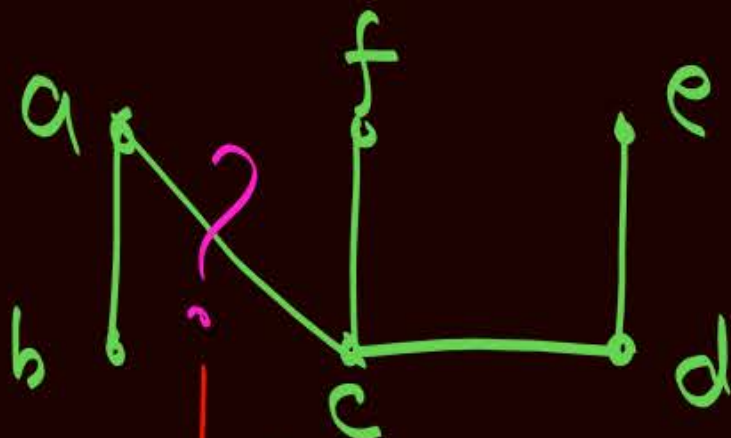






- ' $H_3$ '
- it is a subgraph of  $G$  ✓
  - it is a tree ✓
  - It is acyclic ✓
  - It is connected ✓

But it does not contain all vertices of graph  $G$ ,  $\therefore$  it is not a spanning tree of  $G$ .



- $H_4$
- it is a tree ✓
  - It is acyclic ✓
  - It is connected ✓

Contains all vertices of  $G$  ✓

edge  $\{a,c\} \notin G$ ,  $\therefore H_4$  is not a subgraph of  $G$ . Hence,  $H_4$  is not a spanning tree of  $G$ .



① If graph  $G$  is connected it will have at least '1' spanning tree

② Graph  $G$  will have exactly '1' spanning tree if and only if graph  $G$  itself is a tree.

{ in that case, spanning tree will be graph  $G$  itself }

Note:

Let  $G$  be a connect graph with  $n$ -vertices,  
then each spanning tree of  $G$  will contain  
exactly  $(n-1)$  edges.





## Topic : Circuit Rank

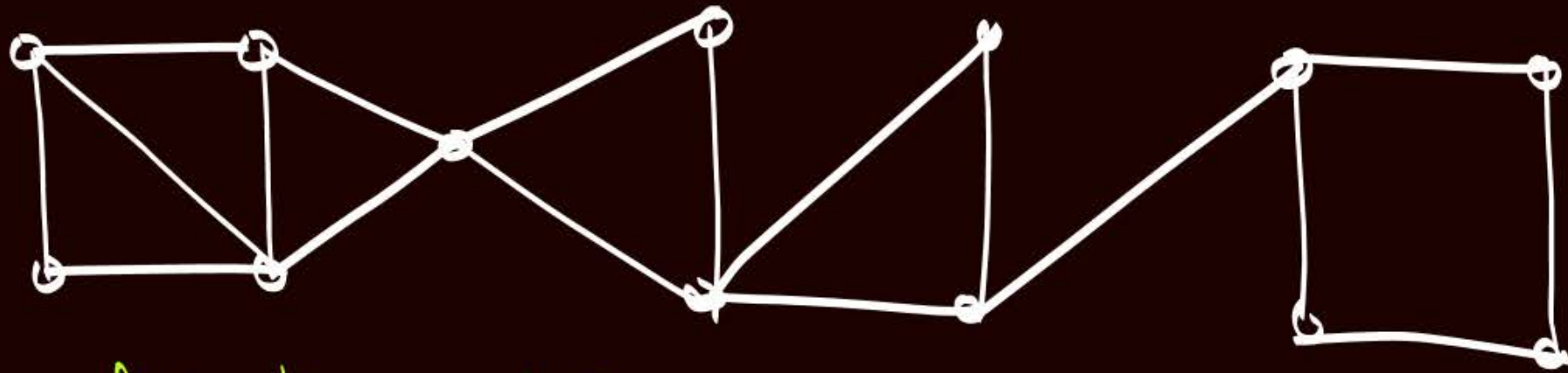
✦ Number of edges that must be deleted from a connected graph  $G$  in order to obtain a spanning tree of  $G$  is called circuit rank of graph  $G$ .

✦ Let  $G$  is a Connected graph with  $n$ -vertices and  $m$ -edges.

$$\begin{aligned} \text{then Circuit Rank of graph } G &= \underbrace{m}_{\substack{\text{No. of Edges} \\ \text{in original} \\ \text{graph}}} - \underbrace{(n-1)}_{\substack{\text{No. of Edges} \\ \text{Present in} \\ \text{Spanning tree}}} = (m-n+1) \end{aligned}$$

eg Find the Circuit rank of the following graph,

$G =$

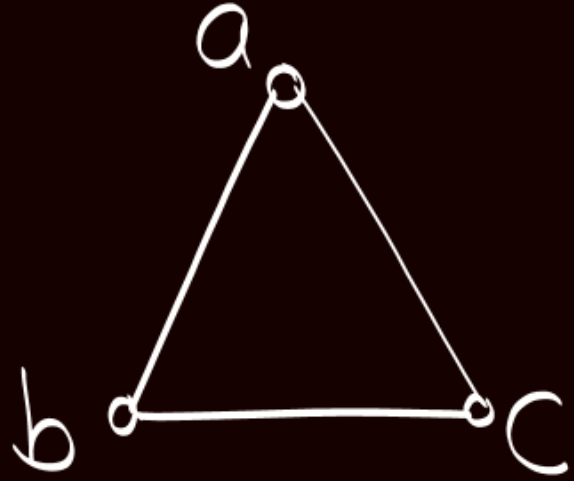


No. of Vertices: 13

No. of Edges in  $G = 18$

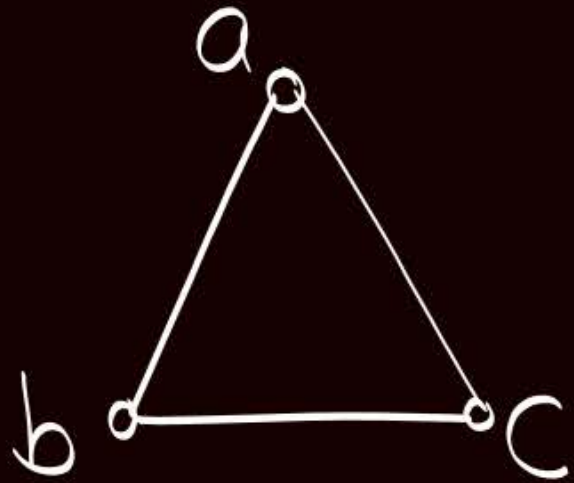
Circuit Rank:  $18 - 12 = 6$

Q:- Find the no. of spanning trees of following graph

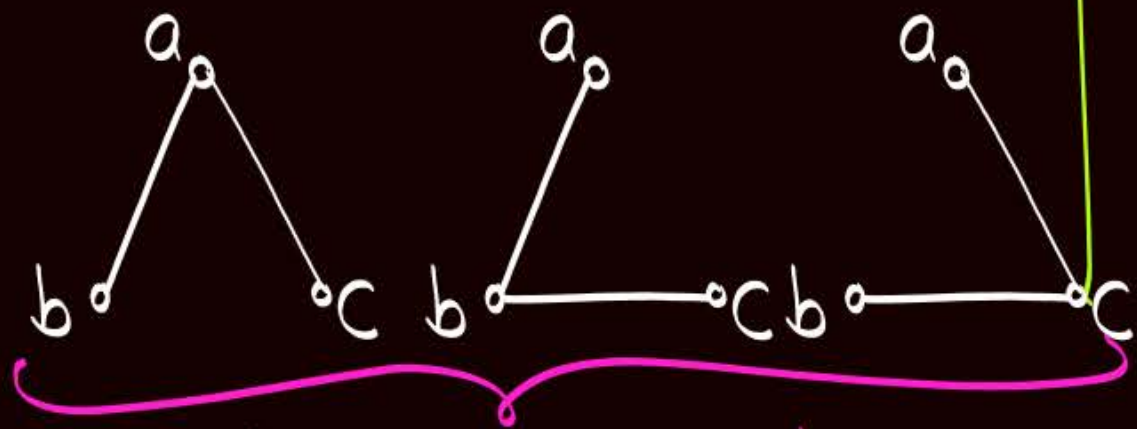




Q:- Find the no. of spanning trees of following graph



$$\text{Circuit Rank} = 3 - 2 = 1$$



'3' spanning trees

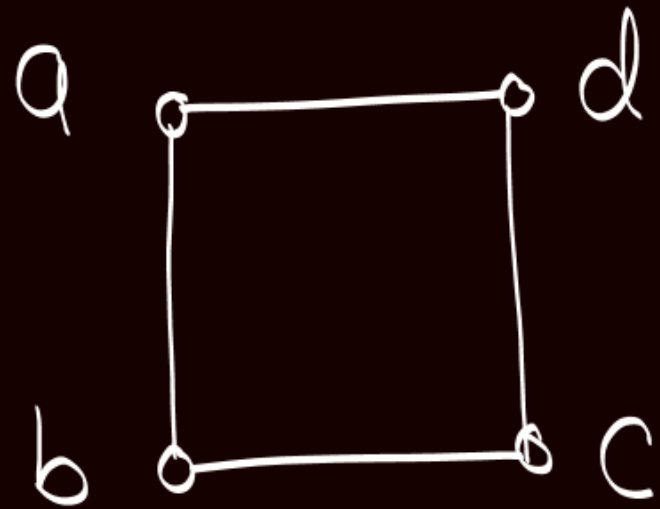
There is a cycle of length = 3

From a cycle of length = 3  
We can delete an edge in  $3C_1$  ways

And

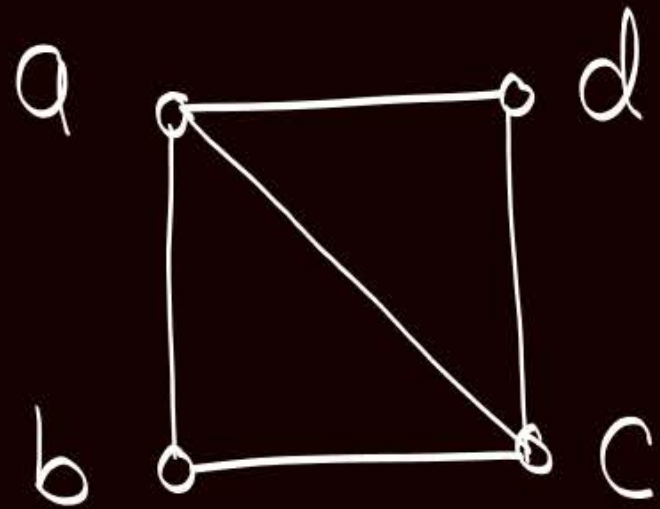
Hence '3' spanning trees are possible

Q:- Find the no. of spanning trees of following graph



Ans = 4

Q:- Find the no. of spanning trees of following graph  
 out of '5' edges we need to delete '2' edges

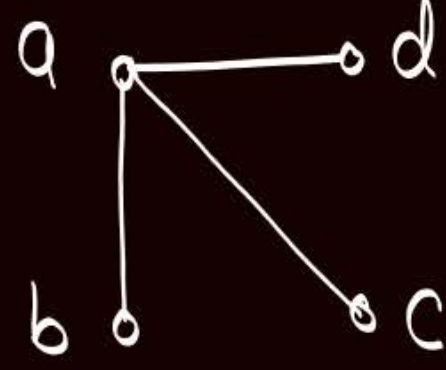
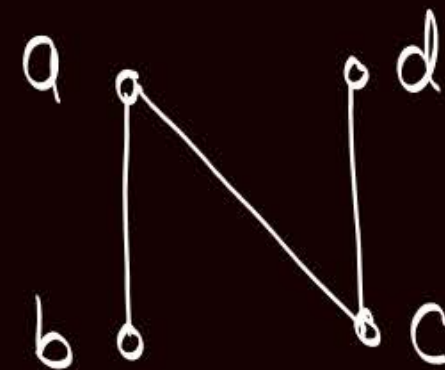
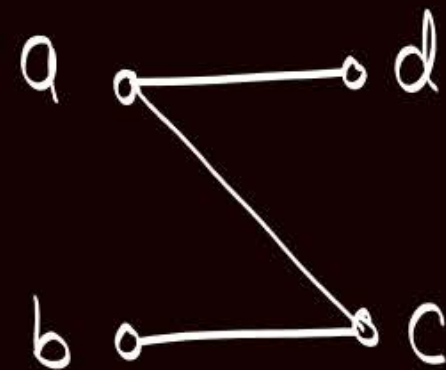
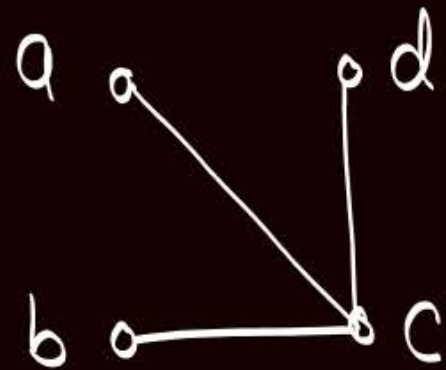
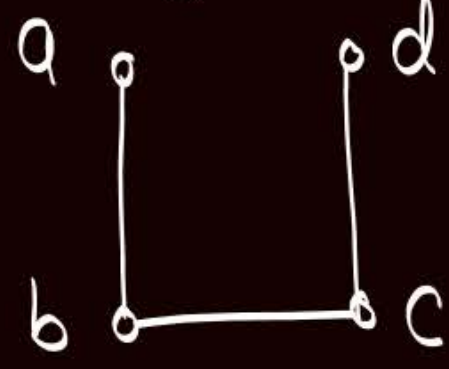
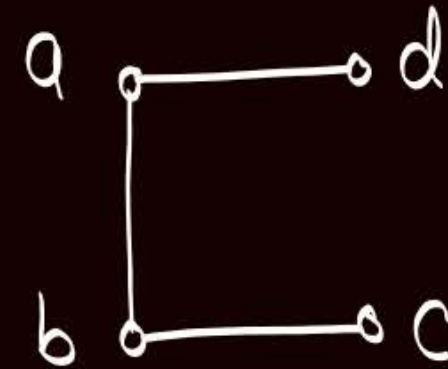
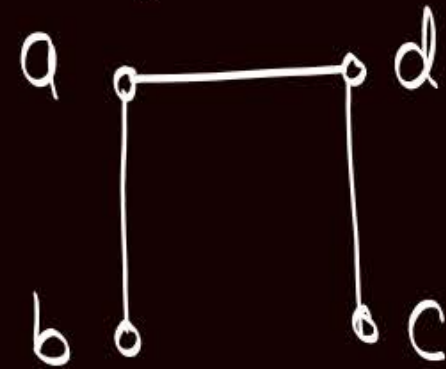
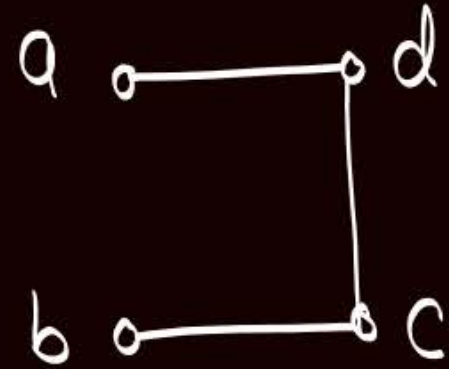
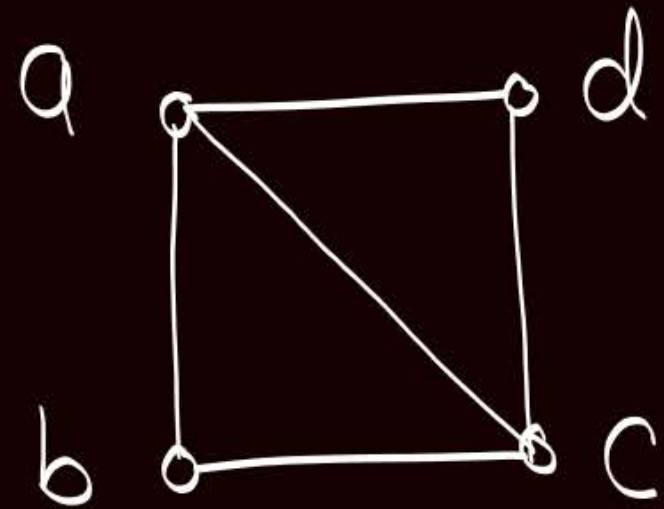


No. of vertices = 4  
 No. of Edges = 5  
 Circuit Rank =  $5 - 3 = 2$

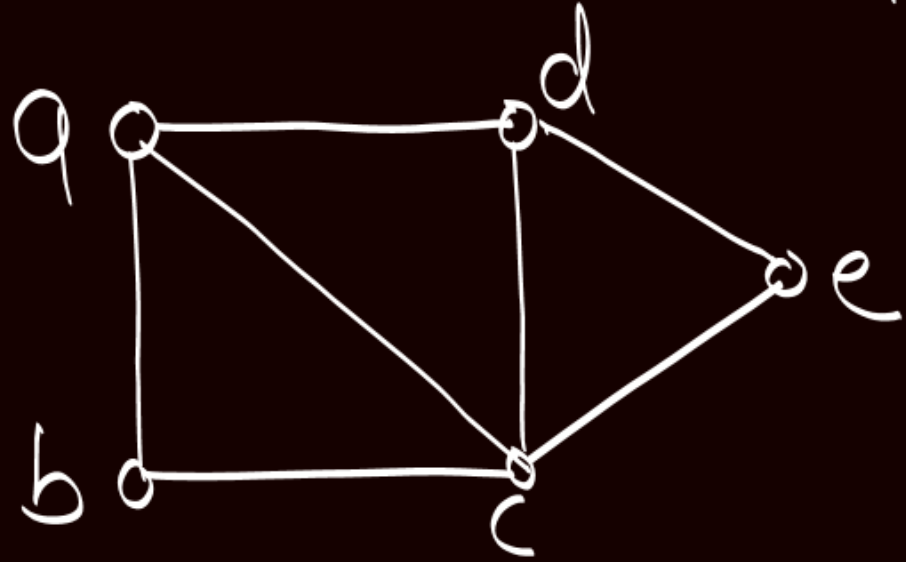
|                             |   |                                                  |   |                                                                            |
|-----------------------------|---|--------------------------------------------------|---|----------------------------------------------------------------------------|
| Total no. of spanning trees | = | Total no. of ways to delete '2' out of '5' edges | — | No. of ways to delete '2' edges such that resulting subgraph is not a tree |
|                             |   | $\downarrow$                                     |   |                                                                            |
|                             |   | $= {}^5C_2$                                      | — | 2                                                                          |
|                             |   | $= 10$                                           | — | 2                                                                          |
|                             |   | $= 8$                                            |   |                                                                            |



Q:- Find the no. of spanning trees of following graph



H.W Q:- Find the no. of spanning trees of following graph



## \* Kirchoff's Matrix Theorem:-

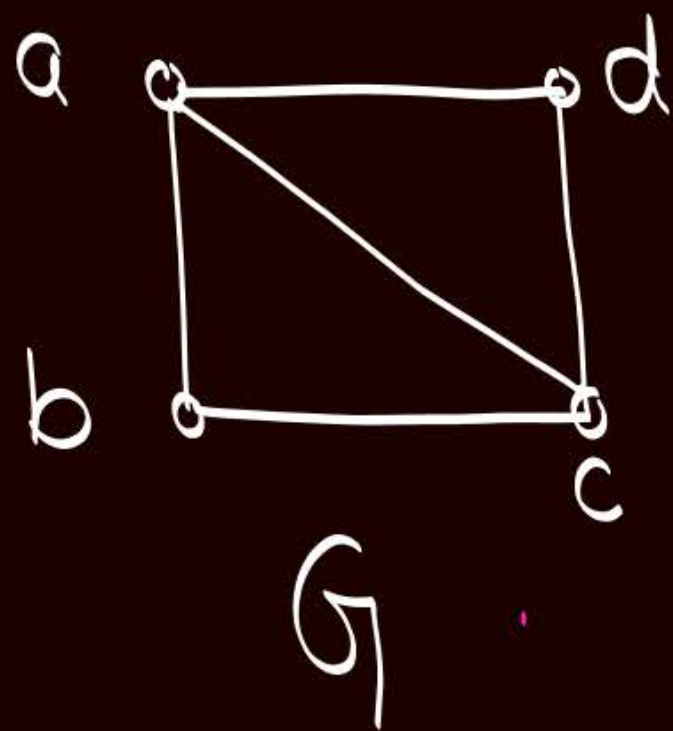
Kirchoff's matrix can be used to find the number of spanning trees of a given graph.



★ Kirchoff's Matrix Theorem:-

Adjacency matrix is a way of representing the graph

Step-1 → Let 'A' is the adjacency matrix w.r.t. given graph



Adjacency matrix  
w.r.t. graph G

$$A = \begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

## \* Kirchoff's Matrix Theorem:-

Step-2 • 'M' is a matrix obtained from adjacency matrix 'A' by replacing all 1's of matrix A by '-1' and by replacing all the 0's of principle diagonal of matrix 'A' by the degrees of corresponding vertices.  
} All other 0's of matrix 'A' will remain as it is }

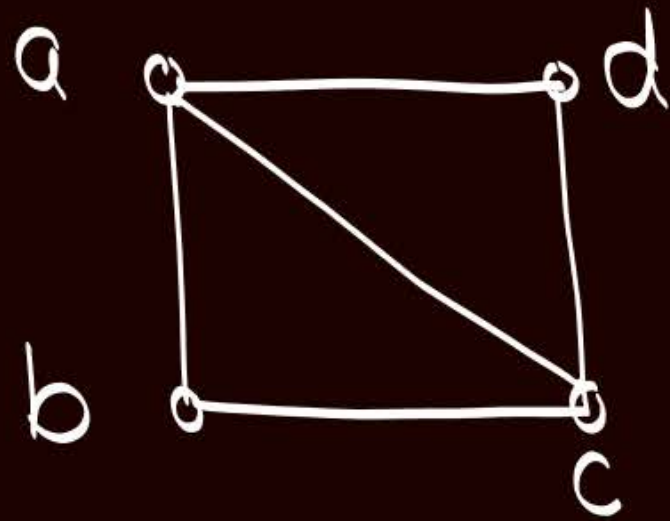
$$M = D - A$$

Degree Matrix  
w.r.t. given graph

Matrix M is called  
Laplacian Matrix



# ★ Kirchoff's Matrix Theorem:-



G

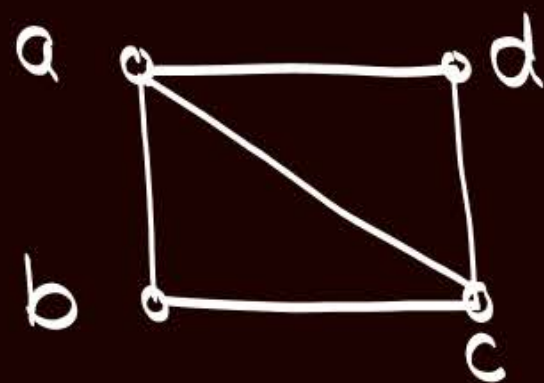
$$A = \begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

$\Rightarrow$

$$M = \begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} 3 & -1 & -1 & -1 \\ -1 & 2 & -1 & 0 \\ -1 & -1 & 3 & -1 \\ -1 & 0 & -1 & 2 \end{bmatrix} \end{matrix}$$



# \* Kirchoff's Matrix Theorem:-



⇒ Degree matrix w.r.t. graph  $G$ :

$$D = \begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix} \end{matrix}$$

$$M = D - A$$

$$M = \begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix} \end{matrix} -$$

$$\begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \end{matrix} A$$

$$= \begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} 3 & -1 & -1 & -1 \\ -1 & 2 & 1 & 0 \\ -1 & 1 & 3 & 1 \\ -1 & 0 & 1 & 2 \end{bmatrix} \end{matrix}$$

\* Laplacian matrix is a symmetric matrix

\* In laplacian matrix, summation of the elements of each row and each column is always "Zero"

∴ Laplacian matrix is a singular matrix  
i.e. determinant value = 0



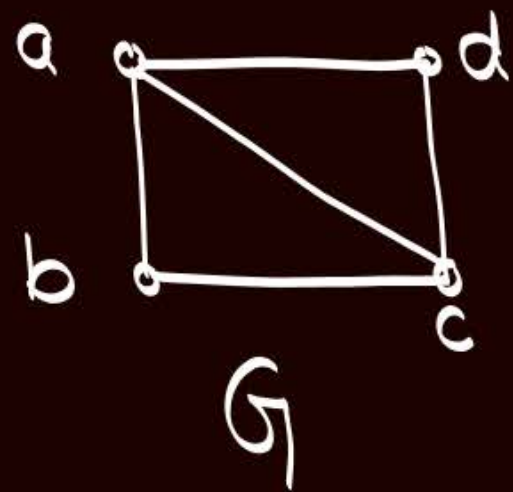
## ★ Kirchoff's Matrix Theorem:-

- Once we have obtained matrix  $M$ ,  
Co-factor w.r.t. any element of matrix  $M$   
will give us the number of spanning trees  
Corresponding to the given graph

Co-factor of  $a_{ij} = (-1)^{i+j} \times$  determinant value while  
skipping elements of  $i^{\text{th}}$  row  
&  $j^{\text{th}}$  Column.



# ★ Kirchhoff's Matrix Theorem:-



$$A = \begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$



$$M = \begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} 3 & -1 & -1 & -1 \\ -1 & 2 & -1 & 0 \\ -1 & 1 & 3 & -1 \\ -1 & 0 & -1 & 2 \end{bmatrix} \end{matrix}$$

*Note: In the original image, the element 3 in the first row, first column of M is labeled a<sub>11</sub>.*

$$\begin{aligned} \text{Co-factor of } a_{11} &= (-1)^{1+1} \begin{vmatrix} 2 & -1 & 0 \\ -1 & 3 & -1 \\ 0 & -1 & 2 \end{vmatrix} = + (2 \times (6-1) + 1(-2)) \\ &= 10 - 2 \\ &= 8 \end{aligned}$$

## \* Kirchoff's Matrix Theorem:-

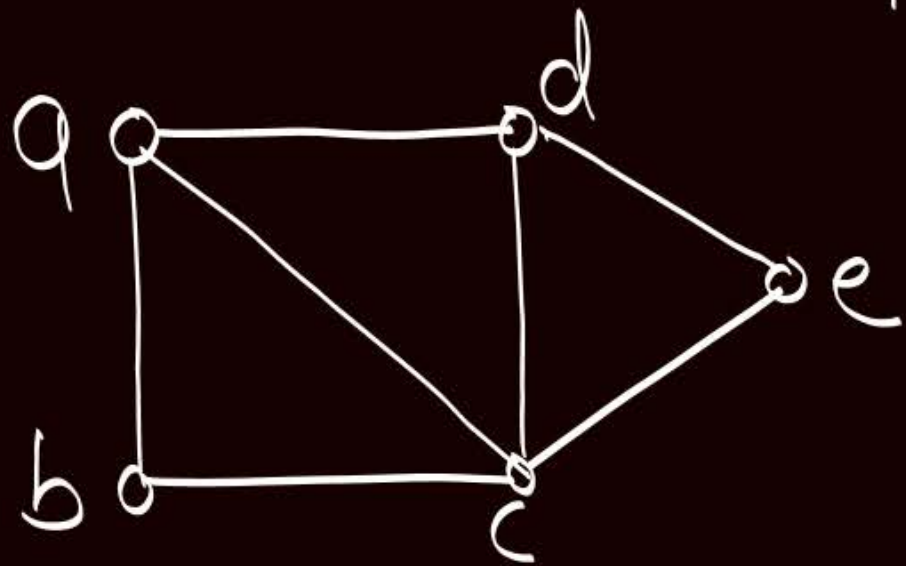
Step 1: Write adjacency matrix 'A' w.r.t. given graph

Step 2: Obtain laplacian matrix 'M' from matrix 'A' by replacing 1's in matrix A by '-1' and by replacing 0's of principle diagonal in matrix 'A' by the degree of corresponding vertex

Step 3: Co-factor w.r.t. any element of matrix 'M' will give us the number of spanning tree



H.W Q:- Find the no. of spanning trees of following graph



$$\Rightarrow M = \begin{matrix} & \begin{matrix} a & b & c & d & e \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \\ e \end{matrix} & \begin{bmatrix} 3 & -1 & -1 & -1 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ -1 & -1 & 4 & -1 & -1 \\ -1 & 0 & -1 & 3 & -1 \\ 0 & 0 & -1 & -1 & 2 \end{bmatrix} \end{matrix}$$

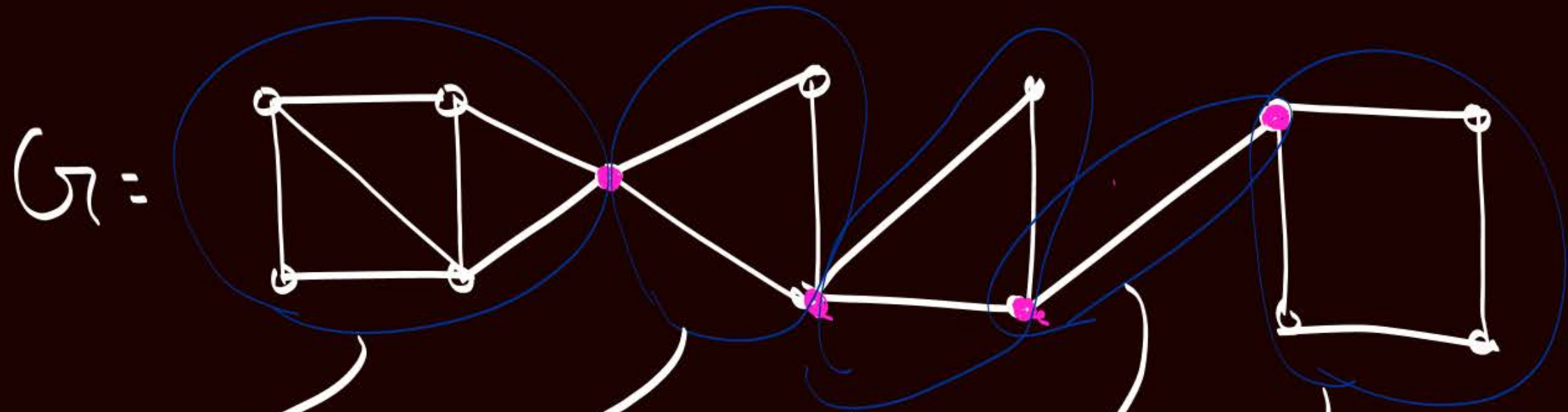
$$\text{Cofactor of } a_{11} = (-1)^{1+1} \begin{vmatrix} 2 & -1 & 0 & 0 \\ -1 & 4 & -1 & -1 \\ 0 & -1 & 3 & -1 \\ 0 & -1 & -1 & 2 \end{vmatrix} = 2 * \begin{vmatrix} 4 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 2 \end{vmatrix} - (-1) \begin{vmatrix} -1 & -1 & -1 \\ 0 & 3 & -1 \\ 0 & -1 & 2 \end{vmatrix}$$

$$= 2 [4(6-1) + 1(-2-1) - 1(1+3)] + [-1(6-1)]$$

$$= 2 [20 - 3 - 4] + (-5) = 26 - 5 = 21$$



eg Find the number of spanning trees of the following graph



Total no.  
of spanning  
trees =

$$21 * 3 * 3 * 1 * 4 = \underline{\underline{756}}$$



## Topic : Caley's Formula

\* For a Complete graph  $K_n$   
number of spanning trees can be given by  $= n^{(n-2)}$

Gate-2024

How many spanning trees are possible  
for a complete graph with '4' vertices {i.e.  $K_4$ }

$$\# \text{ Spanning trees of } K_4 = (4)^{(4-2)} = 16$$





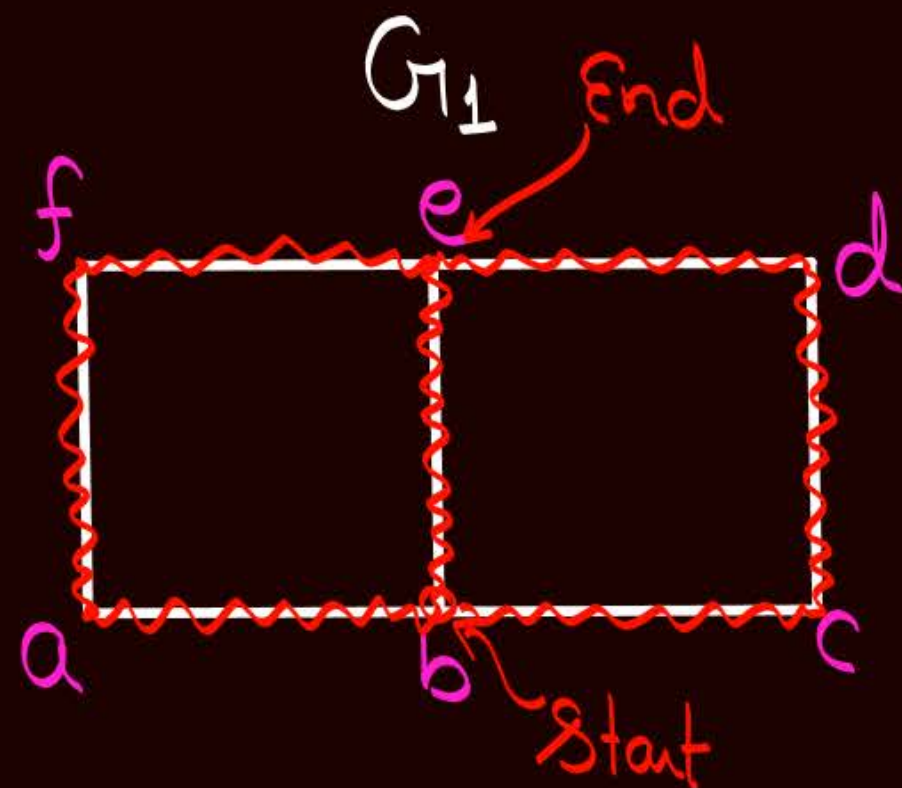
## Topic : Euler trail / Euler path

i.e. Start & End vertex  
are different

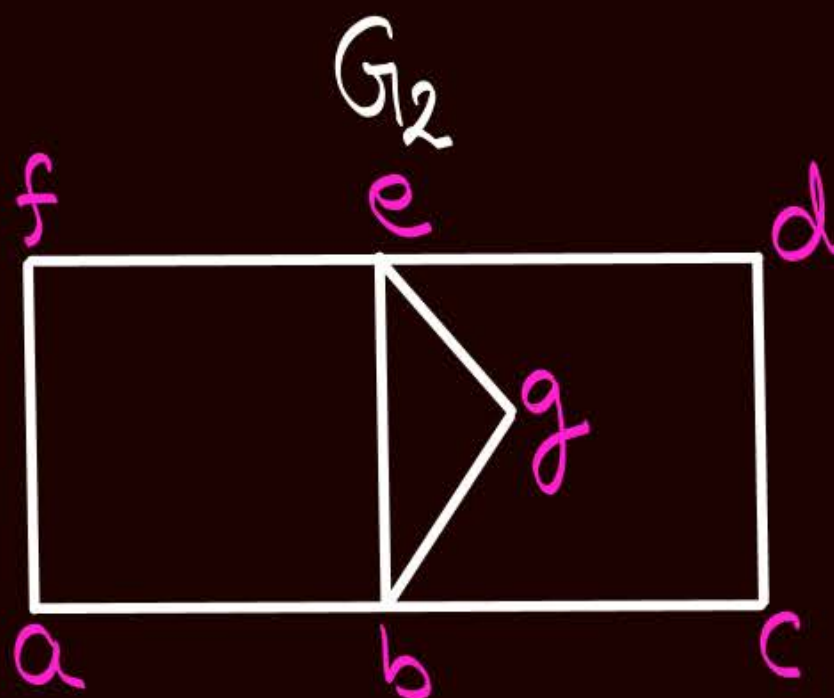


→ Let  $G$  be a Connected graph, an open trail that visit every edge of the graph exactly once and every vertex of the graph at least once is called an Euler trail / Euler path

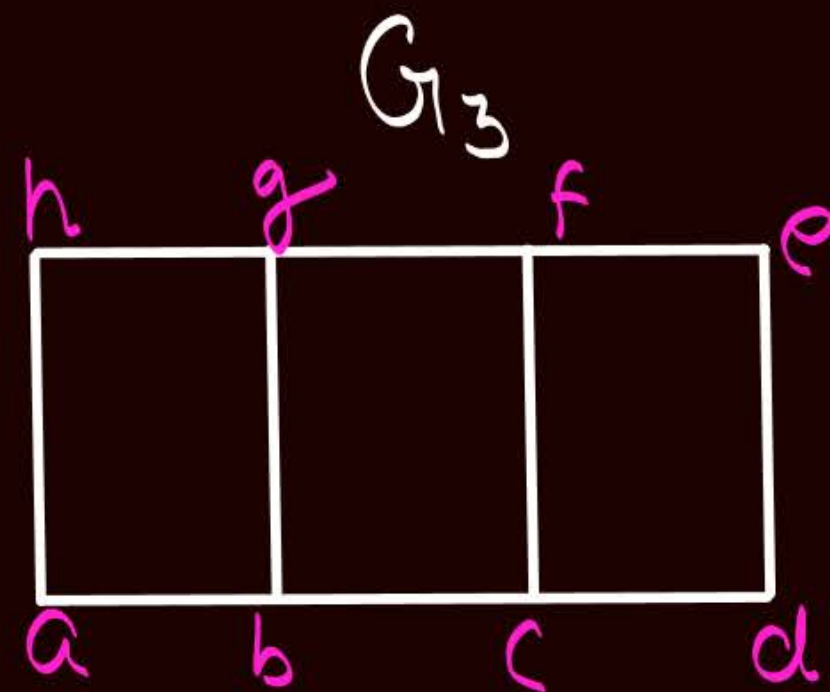




Euler trail (open)  
Exist in graph  $G_1$



Euler trail (open)  
does not  
Exist in graph  $G_2$



Euler trail (open)  
does not  
Exist in graph  $G_3$



## Topic : Euler circuit

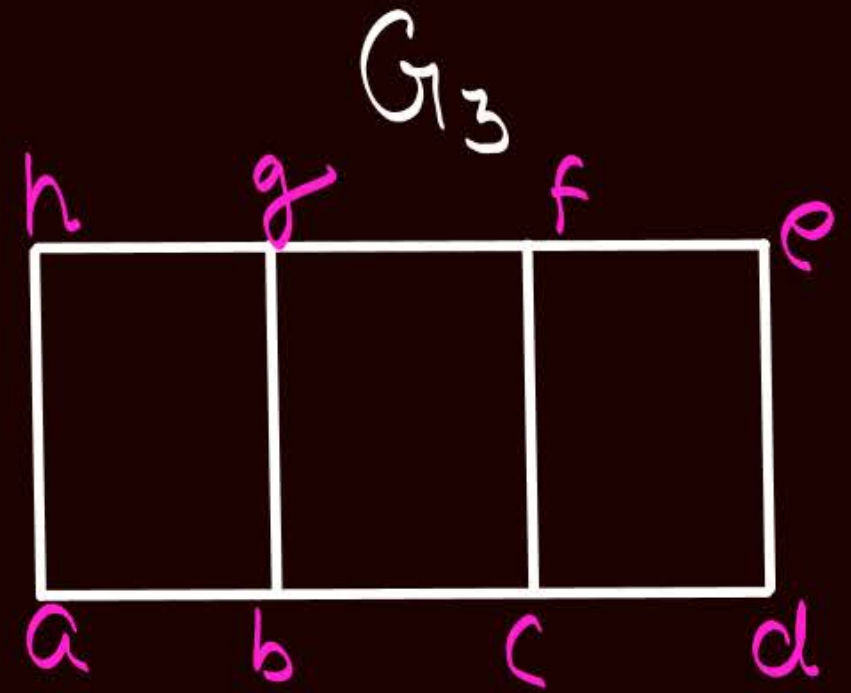
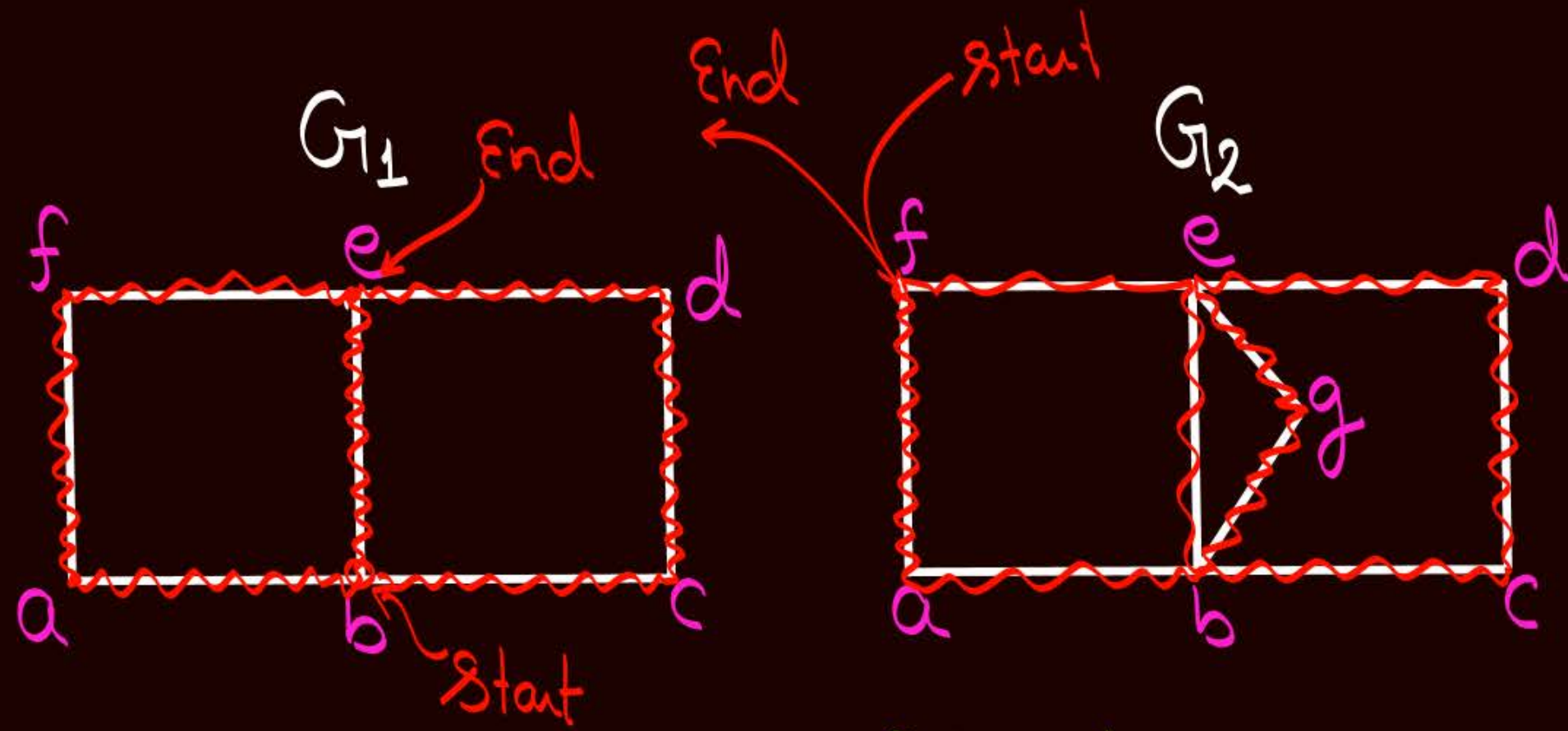


Closed

Start & End vertex  
must be same

In a connected graph  $G$ , a Closed trail that visit Every edge of the graph Exactly once and every vertex of the graph at least once, is called an "Euler Circuit"



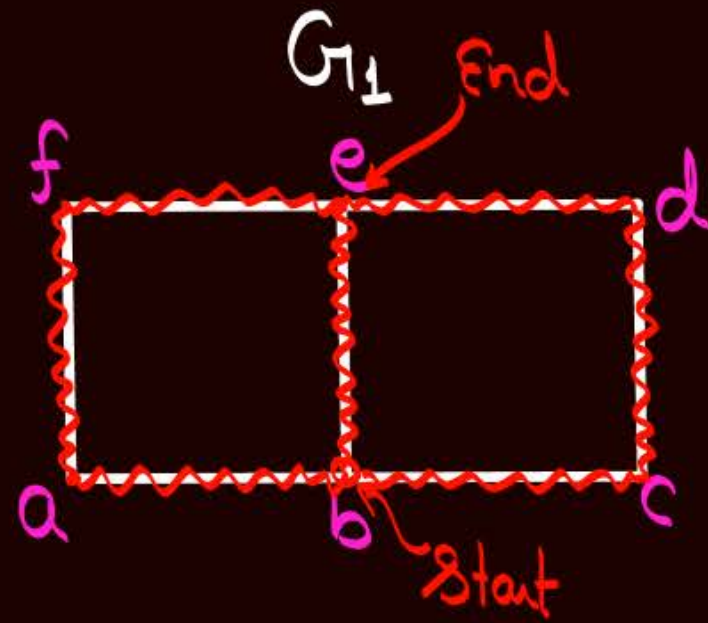


Euler trail (open)  
Exist in graph  $G_1$   
But Euler Circuit  
does not Exist in  $G_1$

Euler trail (open)  
does not  
Exist in graph  $G_2$   
But Euler Circuit  
Exist in  $G_2$

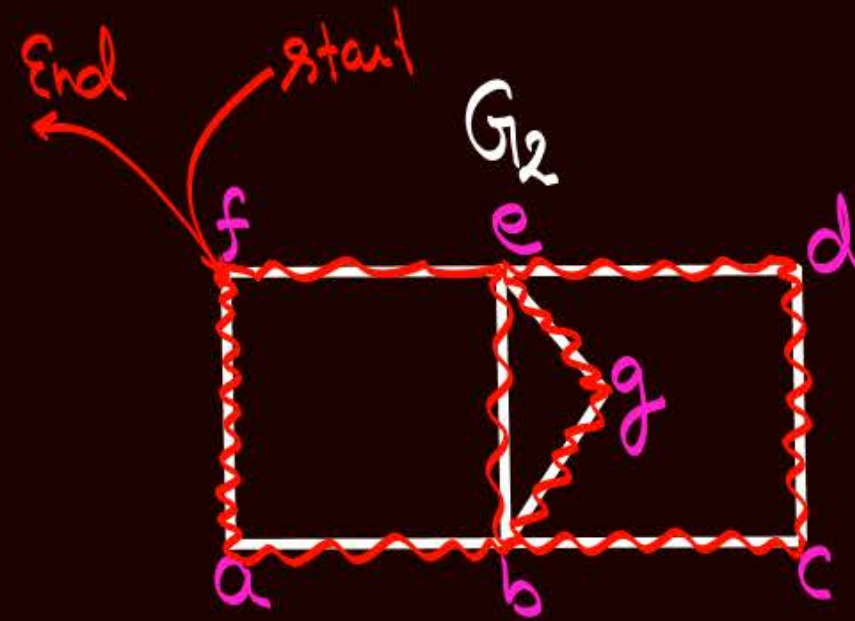
Euler trail (open)  
does not  
Exist in graph  $G_3$   
Euler Circuit also  
does not exist in  $G_3$





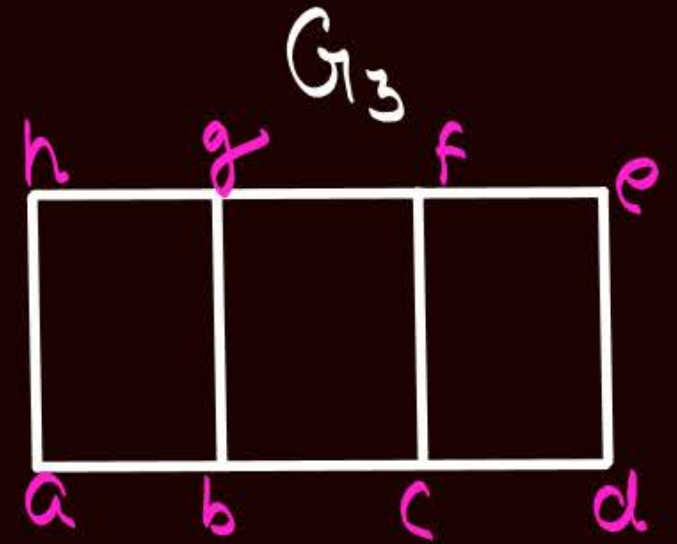
Euler trail (open)  
Exist in graph  $G_1$   
But Euler Circuit  
does not Exist in  $G_1$

In graph  $G_1$  there  
Are exactly two  
Vertices of odd degree



Euler trail (open)  
does not  
Exist in graph  $G_2$   
But Euler Circuit  
Exist in  $G_2$

In graph  $G_2$  there  
Are '0' vertices of  
odd degree



Euler trail (open)  
does not  
Exist in graph  $G_3$   
Euler Circuit also  
does not exist in  $G_3$

In graph  $G_3$  there are  
'4' {i.e. more than '2'} Vertices  
of odd degree



Note:- →

- ① If graph  $G$  is Connected graph, and there are no vertices of odd degree, then Euler Circuit will exist in graph  $G$ , but no Euler trail.
- ② If graph  $G$  is Connected graph, and there are exactly two vertices of odd degree, then Euler Circuit will not exist in graph  $G$ , but Euler trail exist.
- ③ If graph  $G$  is Connected graph, and there are more than two vertices of odd degree, then no Euler Circuit and no Euler trail will exist in  $G$ .

Note:- If graph  $G$  is a disconnected graph, then there will be no Euler trail and no Euler Circuit.

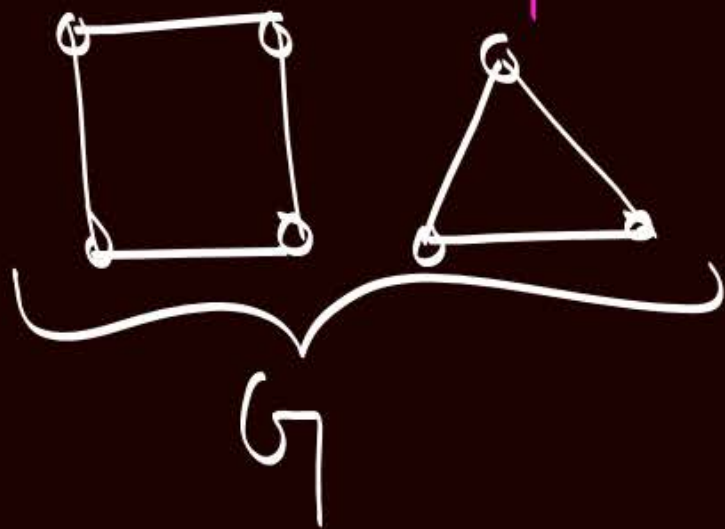


→ Check whether the statement is true or false,

• In a graph  $G$  if all vertices are of even degree then Euler circuit exists in graph  $G$ .

→ False, because they did not specify that the graph is connected.

e.g.



No vertex of odd degree, but no Euler circuit



## Topic : Eulerian graph / Euler graph

{ A graph  $G$  is an Eulerian graph if and only if Euler circuit exists in graph  $G$ .

(or)

{ A graph  $G$  is an Eulerian graph if and only if graph  $G$  is a Connected graph and degree of each vertex is Even.





## Topic : Traversable graph

A graph  $G$  is called traversable if and only if Euler circuit or Euler trail exists in graph  $G$ .

(or)

A graph  $G$  is called traversable if and only if, graph  $G$  is a connected graph and number of vertices with odd degree are '0' or '2'.



## Topic : Hamiltonian path



(open)

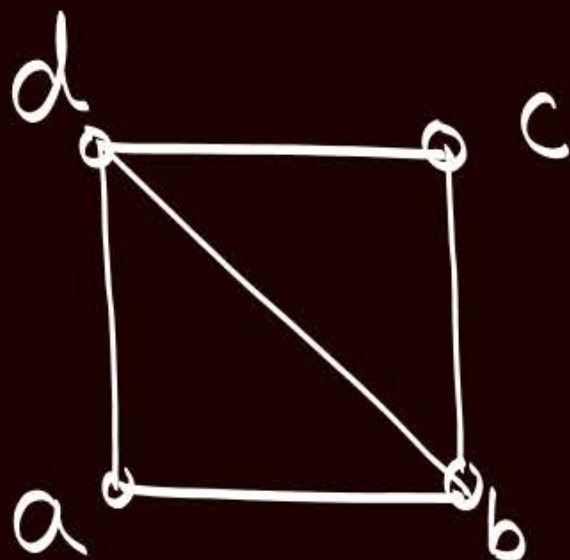
An open path that visit  
Each vertex of the graph Exactly once } we may skip some  
is called Hamiltonian Path } of the edges

(or)

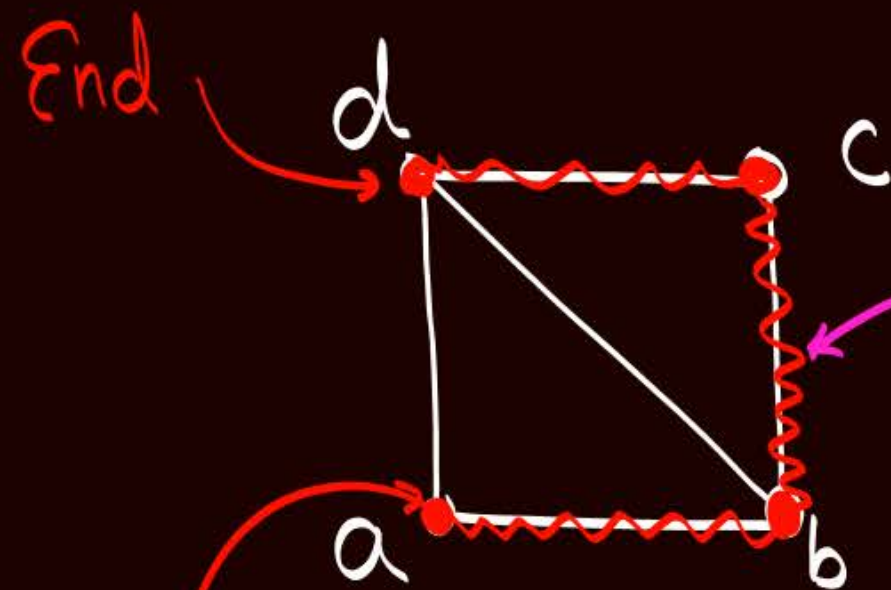
An open path that visit  
Each vertex of the graph Exactly once and Each edge of the graph  
at most once is called Hamiltonian path



eg



$G$



Start

$G$

in this path  
we have visited  
Every vertex  
Exactly once

↓  
∴ Hamiltonian  
Path Exist  
in graph  $G$ .



## Topic : Hamiltonian circuit

(Hamiltonian<sup>(or)</sup> Cycle)

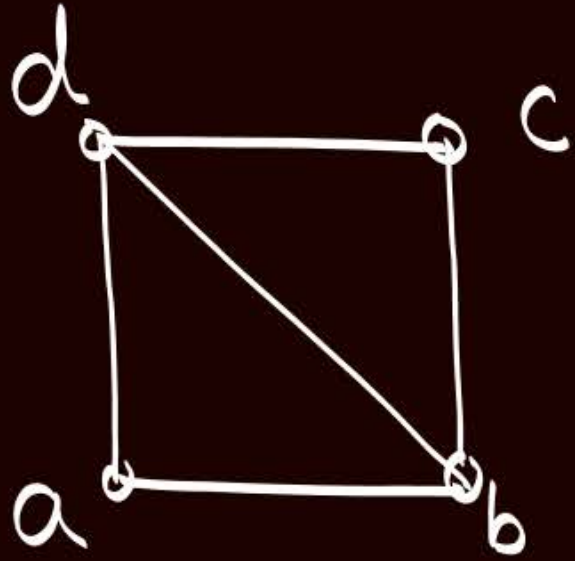
Closed

Start & End vertex  
must be same

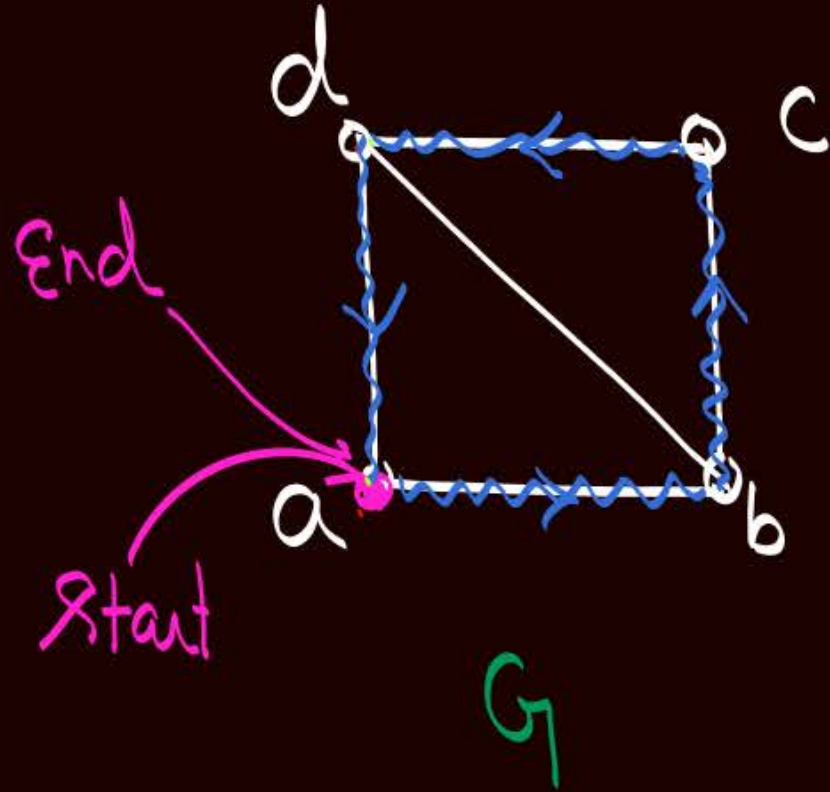
A Closed Path that visit  
Each vertex of the graph Exactly once Except for  
the start vertex {it will be visited twice}, and each edge  
of the graph at most once is called  
Hamiltonian circuit.



eg



$G$

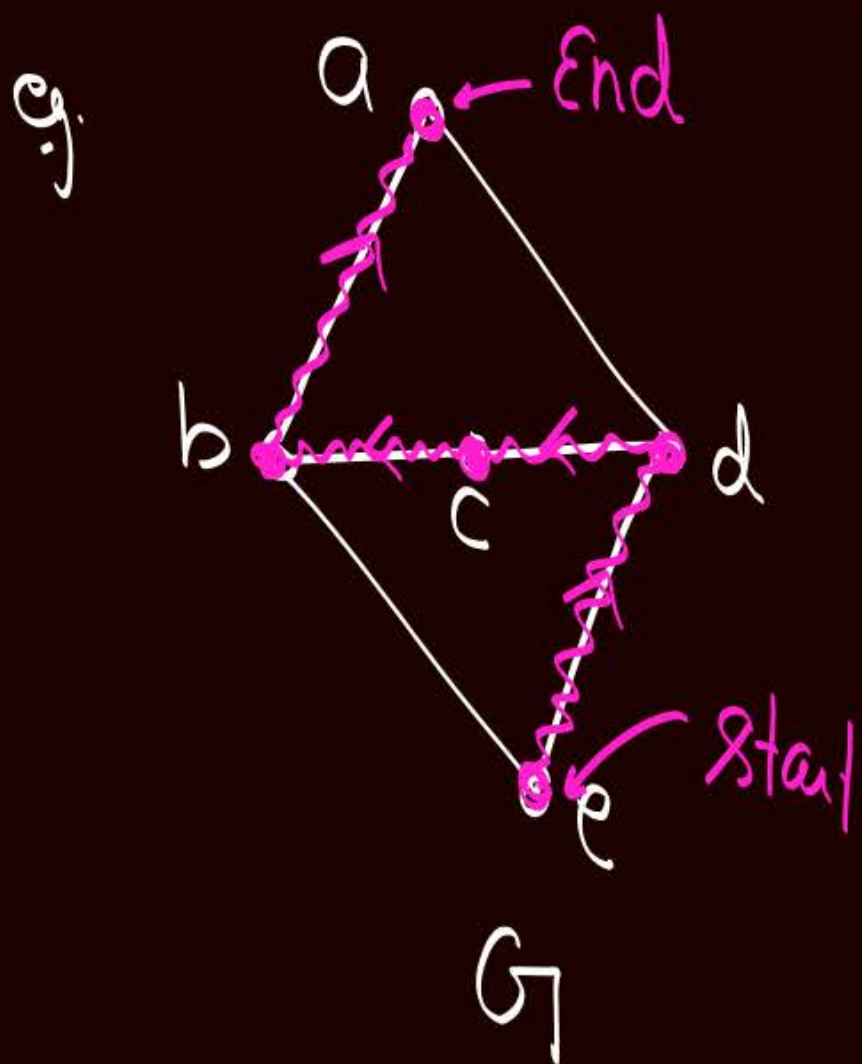


Hamiltonian Circuit  
Exist in graph  $G$

Note :-

If Hamiltonian Circuit exists in the graph  $G$ ,  
then Hamiltonian Path will also exist in graph  $G$ .

But If Hamiltonian Path exist in graph  $G$ ,  
then Hamiltonian Circuit need not exist in graph  $G$   
may or may not



} Hamiltonian path (open) exist  
in graph  $G$ ,  
but Hamiltonian Circuit does not  
exist in graph  $G$ .





## Topic : Hamiltonian graph

A graph  $G$  is called Hamiltonian graph  
if and only if Hamiltonian circuit exist  
in graph  $G$

Note:-

There is no trick to check whether an arbitrary graph is a Hamiltonian graph or not,  $\therefore$  we need to analyse the graph to answer the question.





## Topic : Dirac's theorem

- Let  $G = (V, E)$  be a graph with  $n$  vertices in which each vertex has degree at least  $n/2$ , Then  $G$  has a Hamiltonian cycle.

and hence ' $G$ ' is a Hamiltonian graph



## Topic : Distance



➤ The distance between two vertices of graph  $G$  is defined as number of edges on the shortest path {i.e. the path that contain minimum no. of edges} between those two vertices.



distance  
b/w

$$a-b = 3$$

$$a-c = 4$$

$$a-d = 6$$

$$a-e = 1$$

$$a-f = 2$$

$$a-g = 3$$

$$a-h = 4$$

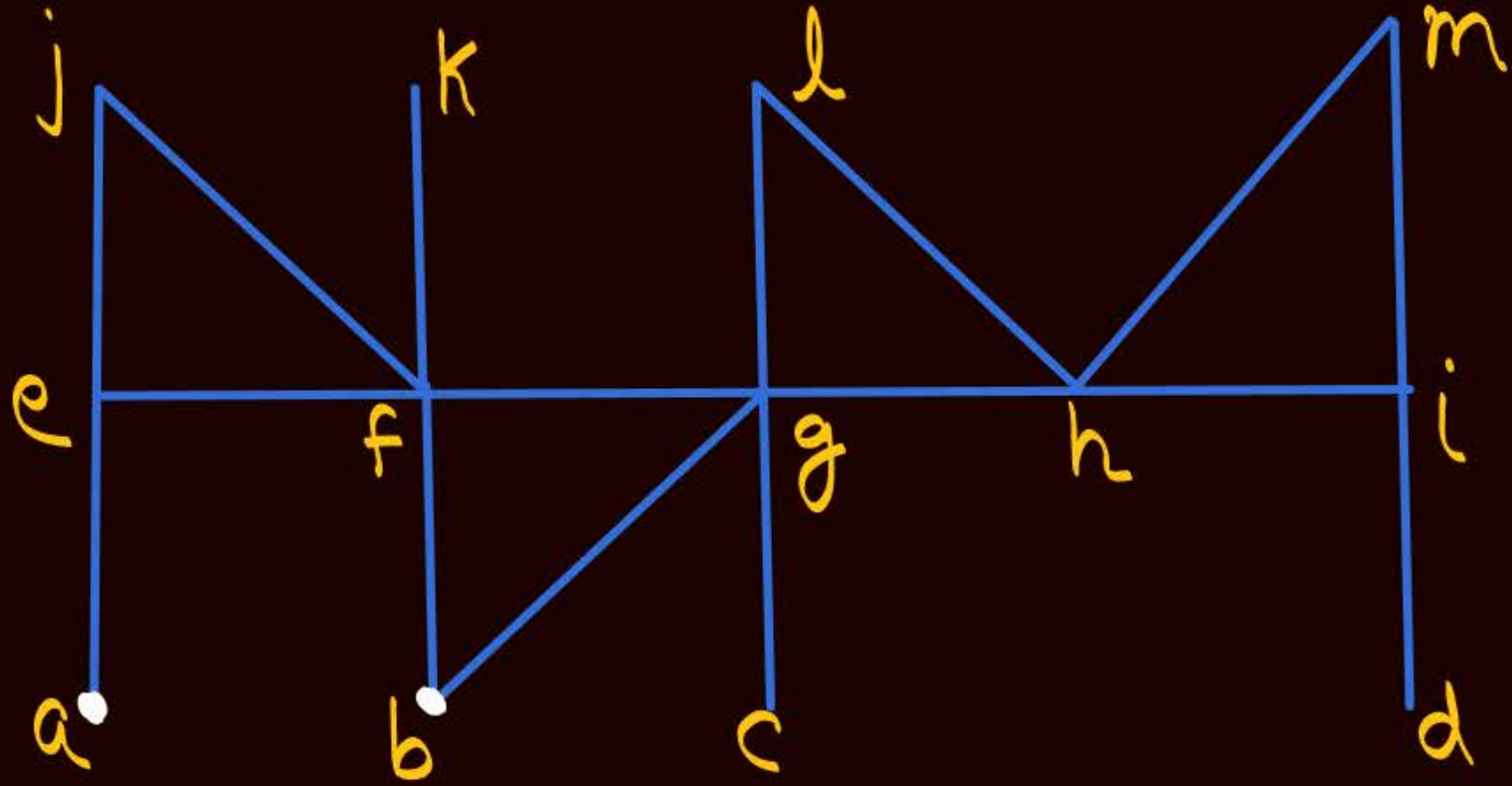
$$a-i = 5$$

$$a-j = 2$$

$$a-k = 3$$

$$a-l = 4$$

$$a-m = 5$$



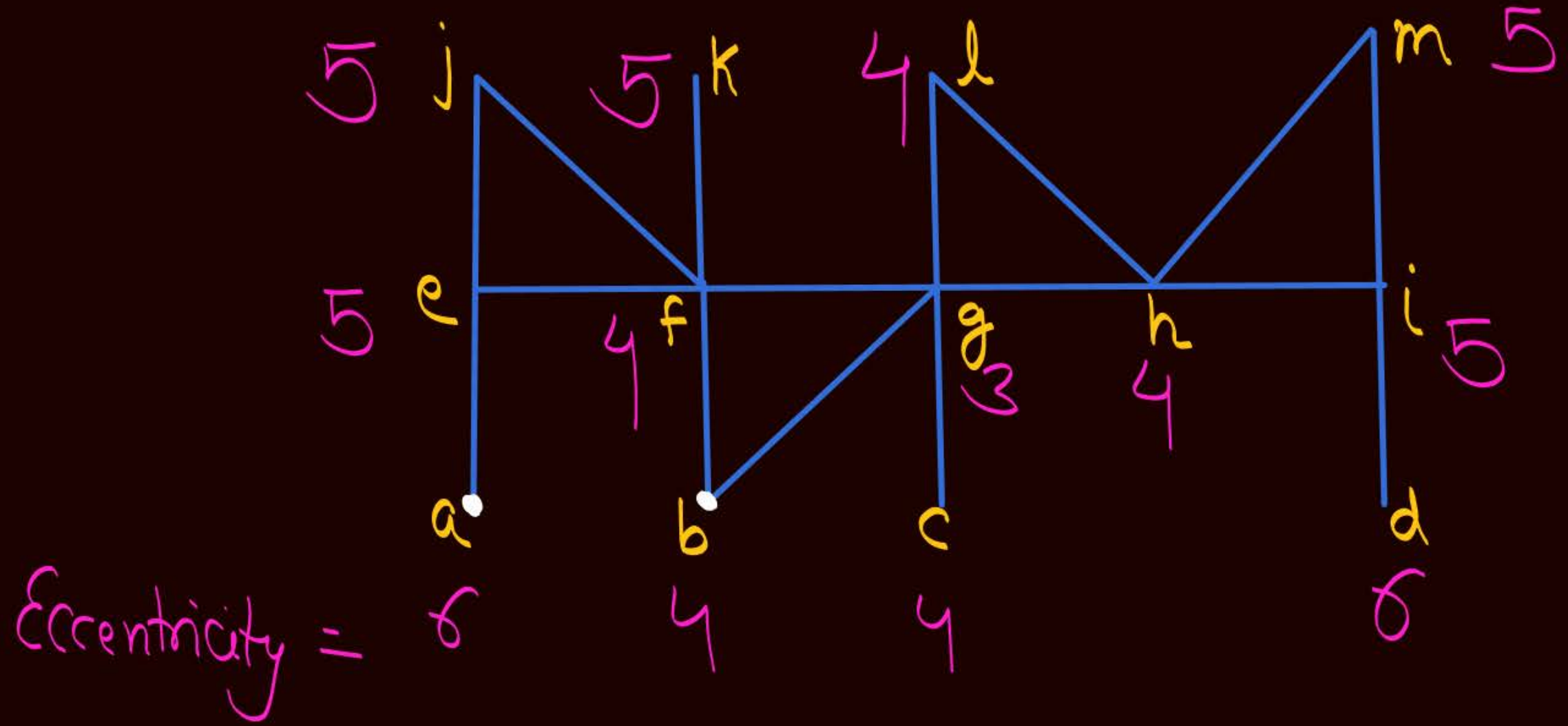


## Topic : Eccentricity



→ Eccentricity of a vertex  $v \in G$ , is the maximum distance of vertex  $v$  from some other vertex of graph  $G$ .







## Topic : Diameter



{ Diameter of graph  $G$  is  
the distance between two  
farthest vertices of graph  $G$  }

➤ Diameter of graph  $G$  is defined as maximum of the eccentricities of all the vertices of graph  $G$ .

\* In the above example,  
diameter of graph  $G = 6$





## Topic : Radius



Radius of graph  $G$  is defined as minimum of the eccentricities of all the vertices of graph.  $G$ .

- In the above example,  
Radius of graph  $G = 3$



## Topic : Centre



→ The set of all vertices with minimum eccentricity is called Centre of the graph.

• In the above example,  
Centre of graph:-  $\{g\}$





## Topic : Centre



→ The set of all vertices with minimum eccentricity is called Centre of the graph.

→ In the above example,

Centre of graph:-  $\{g\}$

→ There may be more than one vertex with minimum eccentricity, in that case Centre will be defined using a set containing all vertices with minimum eccentricity.



## Topic : Girth



In graph theory, the girth of an undirected graph is the length of a shortest cycle contained in the graph. If the graph does not contain any cycles (that is, it is a forest), its girth is defined to be infinity.





## Topic : n-cube



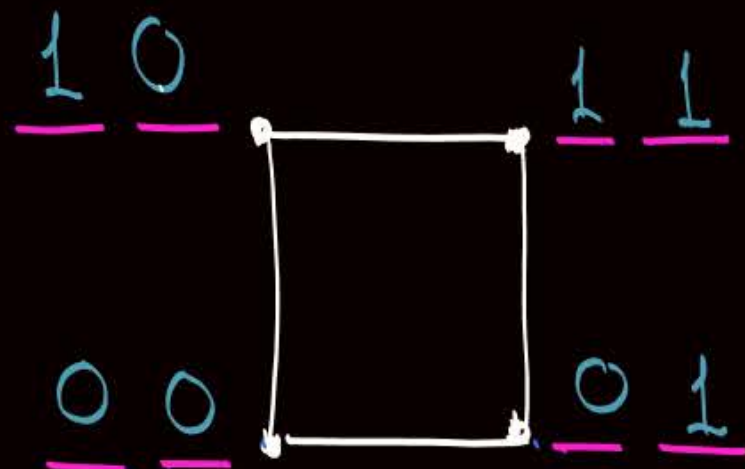
- An n-cube graph, also known as a hypercube graph, is a graph where each vertex is represented by an n-bit binary string. Two vertices are connected by an edge if and only if their corresponding binary strings differ by exactly one bit.

There are  $2^n$  different n-bit binary strings

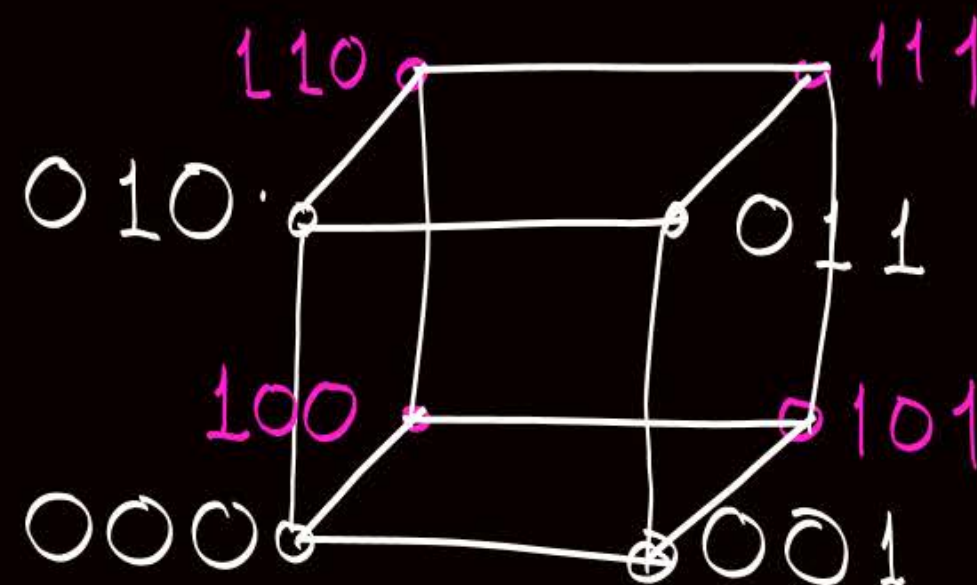
↓  
i.e. we can convert one into another by flipping exactly '1' bit



1-Cube  
 $\downarrow$   
 $2^1 = 2$  vertices



2-Cube  
 $\downarrow$   
 $2^2 = 4$  vertices



3-Cube  
 $\downarrow$   
 $2^3 = 8$  vertices



In a  $n$ -Cube graph,

$$\text{No. of vertices} = 2^n$$

$$\text{No. of Edges} = n \cdot 2^{n-1}$$

← same as boolean algebra

Note: ① Diameter of  $n$ -Cube = " $n$ "

② In an " $n$ -Cube" all cycles are of Even length.  
∴ Every  $n$ -Cube is a bipartite graph



## 2 mins Summary



✓  
Topic

Connectivity

✓  
Topic

Spanning tree

✓  
Topic

Euler trail and Euler circuit

✓  
Topic

Hamiltonian path and Hamiltonian cycle

✓  
Topic

Distance, Eccentricity, Diameter, Radius, Girth

✓  
Topic

n-cube graph



**THANK - YOU**